

Unit 4: Measures of Dispersion

Introduction (concept) –

A measure of central tendency (i.e. average) is a single figure which represents the whole data. It tells us where the centre of data lies, but it does not give any idea about how the observations are distributed around the centre. Two data may have same average but the observations in one data may scattered widely around the centre as compare to other data. Now we try to understand to what extent this single figure (average) is a proper representative of actual values in the data. If most of the values in the data are close to it, then we say that average represents the data properly, otherwise not. c.g. consider the following scores of 3 batsman in 3 innings

Batsman	Score in 3 innings			A.M.
	I	II	III	
A	100	100	100	100
B	100	50	150	100
C	5	195	100	100

The mean score of each of 3 batsmen is same but it does not mean that all of them are equally good. Score of B & C is more scattered around mean as compare to score of A and score of C is more scattered around mean as compare to B. By common sense one can suggest that batsman A is better and consistent and hence more reliable.

Thus only average does not enable us to draw full picture of data. A further description about the degree of scatter ness is necessary for better information about data. " The scatter ness or variation of actual observations in the data from its average is called as dispersion."

The measures used to measure dispersion are known as measures of dispersion.

Objects of measures of dispersion-

- i) To measure reliability of an average.
- ii) To compare the variability of different distributions.
- iii) To control variability.

Requirements of Good measures of dispersion-

- i) It should be simple to understand and easy to calculate.
- ii) It should be clearly and rigidly defined.
- iii) It should be based on all observations.
- iv) It should not be affected by extreme observations.
- v) It should be capable of further algebraic treatments.
- vi) It should have sample stability.

#Types of measures of dispersion-

* Absolute measures – i) Range ii) Quartile Deviation (Q.D.) iii) Mean deviation (M.D.)
iv) Standard Deviation (S.D.)

* Relative measures- i) coefficient of Range ii) coefficient of Q.D. iii) coefficient of M.D.
iv) coefficient of variation (C.V.)

* Differentiate absolute and relative measures of dispersion.-

Absolute measures	Relative measures
i) Absolute measures have a unit in which actual data is measured. ii) It is used to measure degree of variation in the values of data iii) Absolute measures are- Range, Quartile Deviation (Q.D.), Mean deviation (M.D.), Standard Deviation (S.D.)	i) Relative measures are so defined that it is free from unit i.e. it is a pure number. ii) It is used to compare the variability between two or more distributions iii) Relative measures are-coefficient of Range, coefficient of Q.D, coefficient of M.D., coefficient of variation (C.V.)

* **Range-** The difference between largest and smallest values in the data is called range
i.e. Range = $L - S$, where L: largest value, S: smallest value

* **Coefficient of Range-** The ratio of difference between largest and smallest values in the data to the sum of the same is called coefficient of range

i.e. Coefficient of range = $\frac{L - S}{L + S}$, where L: largest value, S: smallest value

Note- In case of continuous frequency distribution takes upper limit of highest class as a largest value and lower limit of lowest class as a smallest value.

Ex1) Find range and coefficient of range for the data 44, 34, 31, 52, 55, 28, 25

Ex2) Find range and coefficient of range for the data given below

X: 21 23 25 34 37 40 42 44

F: 3 8 14 17 12 8 5 2

Ex3) Find range and coefficient of range for the data given below

Class: 6-12 12-18 18-24 24-30 30-36 36-42 42-48

freq.: 11 19 22 29 20 14 9

Merits and demerits of Range-

Merits – i) It is simple to understand and easy to calculate.

ii) It gives quick results and does not required any special knowledge

Demerits-i) It is not based on all observations.

ii) It is not capable of further algebraic treatments.

iii) It has no sampling stability.

iv) It is affected by extreme observations.

v) It is not suitable for open –end classes.

Uses of Range-

Inspite of above limitations range is used in those situations where variations is small and results are required quickly.

i) It is used in controlling quality of industrial products by constructing range chart

ii) It is used to study variation in stock and prices of shares.

iii) It is used for whether forecasts.

* **Standard deviation (S.D.)-** The positive square root of mean of squares of deviation of observations taken from its mean is called standard deviation. It is denoted by σ .

i.e If $X_1, X_2, X_3, \dots, X_n$ be the n observations, its S.D. is given by

$$S.D. = \sqrt{\frac{\sum (X - \bar{X})^2}{n}} \quad \text{where } \bar{X} = \frac{\sum X}{n}$$

Variance(Mean square deviation about mean) – The square of s.d. is called variance

OR The mean of squares of deviation of observations taken from its mean is called standard deviation. It is denoted by σ^2 .

$$\text{i.e var} = (S.D.)^2 = \frac{\sum (X - \bar{X})^2}{n} \quad \text{where } \bar{X} = \frac{\sum X}{n}$$

Coefficient of S.D.- The ratio of S.D. to absolute value of mean is called coefficient of S.D.

$$\text{i.e. coefficient of S.D.} = \frac{S.D.}{|\bar{X}|}$$

Coefficient of variation (c.v.)- The coefficient of S.D. expressed in percentage is known as c.v.
OR The ratio of S.D. to absolute value of mean expressed in percentage is known as c.v.

$$\text{i.e. coefficient of S.D.} = \frac{S.D.}{|\bar{X}|} \times 100$$

Note- c.v. is used to check which of the group out of given groups is consistent (homogeneous)

If $(c.v.)_{\text{group I}} > (c.v.)_{\text{group II}}$ then group II is more consistent or homogeneous

Computations of variance-

Case-I Individual data - let $X_1, X_2, X_3, \dots, X_n$ be the n observations, its variance is obtained by formula

$$\text{Var} = \frac{\sum (X - \bar{X})^2}{n} \quad \text{OR} \quad \text{Var} = \frac{\sum X^2}{n} - (\bar{X})^2 \quad \text{OR} \quad \text{Var} = \frac{\sum X^2}{n} - \left(\frac{\sum X}{n} \right)^2$$

Ex. Find mean, s.d. and c.v. for the data 24, 31, 29, 42, 36, 40, 27, 33

Case-II Discrete data - let $X_1, X_2, X_3, \dots, X_n$ be the n observations having frequencies $f_1, f_2, f_3, \dots, f_n$ respectively its variance is obtained by formula

$$\text{Var} = \frac{\sum f(X - \bar{X})^2}{N} \quad \text{OR} \quad \text{Var} = \frac{\sum fX^2}{N} - (\bar{X})^2 \quad \text{OR} \quad \text{Var} = \frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N} \right)^2$$

Ex. Find mean, s.d. and c.v. for the following data

X: 21 23 26 29 33 36 38 40

F: 5 8 15 18 12 7 3 2

Case-III Discrete data - let classes for the values of variable and there respective frequencies are given, its variance is obtained by formula

$$\text{Var} = \frac{\sum f(M - \bar{X})^2}{N} \quad \text{OR} \quad \text{Var} = \frac{\sum fM^2}{N} - (\bar{X})^2 \quad \text{OR} \quad \text{Var} = \frac{\sum fM^2}{N} - \left(\frac{\sum fM}{N} \right)^2$$

Where M = midpoints of class

Ex. Find mean, s.d. and c.v. for the data given below

Class: 15-25 25-35 35-45 45-55 55-65 65-75 75-85

F: 6 13 23 27 18 10 4

#Properties of S.D.(Variance) -

1) Mean square deviation is minimum if it is taken about mean.

$$\text{i.e.} \quad \frac{\sum (X - A)^2}{n} \geq \frac{\sum (X - \bar{X})^2}{n} \quad \text{where } A \text{ is any constant}$$

2) Effect of change of origin and scale on variance- Variance is independent of change of origin but depends on change of scale

$$\text{i.e. } \sigma_v^2 = (B)^2 \cdot \sigma_u^2 \quad \text{where } u = \frac{x - A}{B}$$

3) Combined variance- Let n_1 and n_2 be the sizes of two groups of observations having means $\bar{X}_1$ and $\bar{X}_2$ and variances σ_1^2 and σ_2^2 respectively, the variance of combined groups is given by

$$\text{Combined var} = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$$\text{Where } d_1^2 = (\bar{X}_1 - \bar{X})^2, d_2^2 = (\bar{X}_2 - \bar{X})^2 \text{ and } \bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

Merits and demerits of S.D.

Merits-

- i) It is clearly and rigidly defined.
- ii) It is based on all observations
- iii) It is capable of further algebraic treatment.
- iv) It has sampling stability.
- v) It is more accurate and reliable measure of dispersion.

Demerits-

- i) It is difficult to understand and to calculate.
- ii) It is affected by extreme observations.
- iii) It is not suitable for open-end classes.
- iv) It is not calculated for qualitative data.

Assignment No.-4

- 1.) What is measure of dispersion? OR Explain concept of measures of dispersion.
- 2.) Differentiate absolute and relative measures of dispersion.
- 3.) Define range, coefficient of range and state merits and demerits of it.
- 4.) Define Q.D., coefficient of Q.D. and state merits and demerits of it.
- 5.) Define variance, S.D., C.V. and state merits and demerits of it.

Choose most correct alternative

1. The measure of dispersion that free from unit in which data is measured is
 a) range b) s.d. c) Q.D. d) c.v.
2. The most reliable measure of dispersion is
 a) range b) s.d. c) Q.D. d) all of these
3. The measure of dispersion that not affected by extreme observations is.....
 a) range b) s.d. c) Q.D. d) c.v.
4. The measure of dispersion that can be determined graphically is
 a) range b) s.d. c) Q.D. d) c.v.
5. The measure of dispersion that does not satisfies most of the criterion of good measures of dispersion is
 a) range b) s.d. c) Q.D. d) c.v.
6. If all the observations in the data are equal, the dispersion is
 a) zero b) any +ve real number c) can't say anything d) none of these
7. If each observation is increased by 4 then measure of dispersion becomes.....
 a) as it is b) increased by 4 c) increased by 16 d) none of these
8. The range of 20 values is 30 . if each value is doubled then range becomes.....

a) 40 b) 60 c) 20 d) 30

9. The A.m. of two groups is same but their S.D. are 24 & 34 respectively, which group is more homogeneous

a) First b) second c) both d) none of these

10. The difference between upper and lower quartiles is 48 and their sum is 100, then Q.D. becomes

a) 24 b) 50 c) 0.48 d) none of these

11. Measure of dispersion that based on extreme observations only is

a) range b) s.d. c) Q.D. d) all of these

12. The measure of dispersion that based on all observations is

a) range b) s.d. c) Q.D. d) all of these

13. Measures of dispersion are affected by change in

a) origin only b) scale only c) origin and scale both d) none of these

14. The c.v. of group A is less than the c.v. of group B, then

a) group A is more homogeneous b) group B is more heterogeneous.

c) (a) and (b) both. d) none of these

15. Measure of dispersion that capable of further algebraic treatment is

a) range b) s.d. c) Q.D. d) all of these

Ex.1. Given: $n=10$, $\sum X=132$, $\sum X^2=2120$ find c.v.

Ex.2. Given: $n=10$, $\sum X=140$, $\sum (X - A.M.)^2=1305$ find c.v.

Ex.3. Given: $n=10$, $\sum X=1125$, $\sum (X - 12.5)^2=212$ find c.v.

Ex.4. Given: $n=10$, $\sum (X - 13)=76$, $\sum (X - 13)^2=1305$ find c.v.

Ex.5. Given: $n=10$, $\sum u=17$, $\sum u^2=87$, where $u=(X - 5)/7$ find c.v. of X

Ex.6. Given: $\sum f=40$, $\sum fX=1205$, $\sum fX^2=95024$ find c.v.

Ex.7. Given: $\sum f=30$, $\sum fu=24$, $\sum fu^2=104$, where $u=(X - 10)/3$ find c.v. of X

Ex.8. The mean and s.d. of 50 observations were found to be 35 and 21.5 respectively. While cross verification it was observed that 2 observations 38 and 25 were taken instead of 48 and 35. Find correct mean and s.d.

Ex.9. The A.M and S.D. of marks of 60 students in division-A are 56 and 45.6 and that of for 70 students in division-B are 63 and 51.6 respectively.

Find i) The division having less variation in marks.

ii) Combined c.v. of marks.

Ex.10. The A.m. score of batsman A and B in 10 innings are 56 and 62. The s.d. of their score are 21.5 and 31.4 respectively. Find which batsman is more consistent? and who is more run getter?

Ex.11. Following is the information about wages of workers in factory A and B.

	Factory -A	Factory-B
No. of workers	80	120
A.M. of wages	625	595

Variance of wages	81.5	90.14
-------------------	------	-------

Find i) Which factory has large wage bill?

ii) which factory has more variation in wages of workers?